ZERO2CODE WEEK-1

Web Basics & Introduction to C

Course Details

TUTOR: Anshad
WEEK: 1
TOPICS: Web Basics, Intro to C, Variables, Control Statements, Loops (Part 1)

Student Details

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BATCH::

DAY - 3 TASKS – Mastering Loops & Decision Making

🔄 Loops in C

Q1. Write a program using a for loop to print numbers from 1 to 10.

✅ **Example Output:**

```
1 2 3 4 5 6 7 8 9 10
```

📝**SOLUTION:**

```
#include <stdio.h>

int main() {

for (int i=1;i<=10;i++){

printf("%d",i);

printf("\t");}

return 0;

}
```

Q2. Modify the program to print numbers in reverse (10 to 1).

✅ **Example Output:**

```
10 9 8 7 6 5 4 3 2 1
```

📝**SOLUTION:**

```
#include <stdio.h>
int main() {
for (int i=10;i>=1;i--){
printf("%d",i);
printf("\t");}
return 0;
}
```

Q3. Write a for loop to print all even numbers between 1 and 20.

✅ **Example Output:**

```
2 4 6 8 10 12 14 16 18 20
```

📝**SOLUTION:**

```
#include <stdio.h>
int main() { for (int i=2;i<=20;i+=2){
printf("\t%d",i);
}
return 0;
}
```

Q4. Write a program that prints this pattern using loops:

```
1
12
123
1234
12345
```

📝**SOLUTION:**

```
#include <stdio.h>

int main() {

for (int i=1;i<=5;i++){

for(int j=1;j<=i;j++)

printf("%d",j);

printf("\n");

} return 0;

}
```

Q5. Write a program that asks for a number (1-7) and prints the corresponding day of the week using ‘if’ condition.

✅ **Example Output:**

```
Enter a number (1-7): 3
Wednesday
```

📝**SOLUTION:**

```
#include <stdio.h>

int main() {

int num;

printf("Enter any number between 1-7:");

scanf(" %d", &num);

if(num== 1) {

printf("%d is Monday\n",num);}

else if(num== 2) {

printf("%d is Tuesday\n",num);}

else if(num== 3) {

printf("%d is Wednesday \n",num);}

else if(num== 4) {

printf("%d is Thursday\n",num);}
else if (num== 5) {
printf("%d is Friday\n",num);}
else if (num== 6) {
printf("%d is Saturday\n",num);}
else if (num== 7) {
printf("%d is Sunday\n",num);}
else{printf("None");
}
return 0;
}
```

Q6. Modify the program to ask the user for a character input and print whether it’s a vowel or a consonant.

✅ **Example Output:**

```
Enter a letter: A
A is a vowel.
```

📝**SOLUTION:**

```
#include <stdio.h>
   int main() {
  char ch;
  printf("Enter any letter");
  scanf(" %c", &ch);
  if(ch=='a') {
  printf("%c is a vowel\n",ch);}
  else if(ch=='e') {
  printf("%c is a vowel\n",ch);}
  else if(ch=='i') {
  printf("%c is a vowel\n",ch);}
  else if(ch=='o') {
```

```
printf("%c is a vowel\n",ch);}

else if (ch=='u') {

printf("%c is a vowel\n",ch);}

else{printf("%c is a consonant",ch);

}

return 0;

}
```

🌐 Web Basics – Reinforcing Concepts

Q7. What is a Web Browser & Search Engine?

✅ **Explain how they are different.**

📝**SOLUTION:**

Software used to access and view a webpage or website is a web browser. Eg:-Google chrome,Mozilla firefox,Safari,Microsoft edge etc.
Whereas Search engine is a website which helps to find out the specific information about a topic,like the best school near me and it will collect all information regarding that. Eg:-Google,Yahoo,Guruji etc.

Q8. What is the role of DNS in the internet?

✅ **Briefly explain how domain names are converted into IP addresses.**

📝**SOLUTION:**

The role of the Domain name system is it helps to convert website names to IP addresses. Computers require IP addresses to find out the website. When we type a website name like www.google.com. DNS will find out its IP address and helps to direct our computer to the correct server.